Compare beams under the same load

Task 2.4–2.5 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Sketch both supports, the span and a fixed ruler reference at the center. Deflection is the change between unloaded and loaded position.

Measure each sample’s span, width, and thickness. Record differences that affect the comparison.

Sample	Span	Width × thickness	Units / material
A			
B			
C			

02 Plan it on paper

Draw the setup with the loading point and a reference line for deflection.

My prediction, chosen feature or design limit

Compare beams under the same load

Task 2.4–2.5 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Load the samples one increment at a time. Measure center deflection at a common load. Unload and inspect. Compare which sample bent and recovered, which kept a bend, and which failed. Identify the limits of the comparison.

Sample	Common load and unit	Center deflection	Recovery after unloading
A			
B			
C			

04 Use the reading

Ch. 13 | Home Sweet Home | printed pp. 111-122
Ch. 14 | From the Ground Up | printed pp. 123-131
Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Recommend a member for one part of your model. Cite a common-load measurement and describe the material response.

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.